Mihir Khatri

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Professional Summary

Undergraduate Computer Science student focused on machine learning systems, scalable AI infrastructure, and distributed computing. Interested in building efficient ML pipelines, GPU-aware inference systems, and production-oriented AI solutions spanning deep learning, MLOps, and large-scale model deployment. Experienced with PyTorch, CUDA, transformer architectures, and optimisation of memory-constrained inference workloads through independent research and hands-on engineering projects.

Education

Sardar Patel Institute of Technology, Mumbai University Mumbai, India
B.Tech in Computer Science and Engineering Expected Aug 2028
GPA: 9.35 / 10.0

Relevant Coursework: Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Discrete Mathematics, Operating Systems, Computer Networks

Languages: Python, C, C++, Java, SQL, CUDA C, JavaScript

Publication

Ro-SVD: Position-Aware SVD Compression of KV Caches in Grouped-Query Attention Models
Zenodo, May 2026

- Proposed online activation-level KV cache compression using position-aware truncated SVD on pre-RoPE key tensors in Grouped-Query Attention architectures.
- Evaluated across 1,602 samples on all 16 LongBench subsets using Qwen2.5-7B-Instruct with 4-bit NF4 quantisation on dual NVIDIA T4 GPUs.
- Achieved 73.91% win rate on ROUGE-L against uncompressed baseline while reducing KV-cache memory usage by 35.1%.
- Designed adaptive staircase rank allocation with per-head energy thresholding to preserve quality in high-entropy attention heads.
- Implemented custom CUDA de-rotation kernel with cuBLAS SGEMM batched low-rank projection.
- Derived randomised SVD optimisation path reducing complexity from $O(Thd^2)$ to $O(Thr \log r)$.

Research & Projects

LLM Inference Optimisation: KV Cache via Low-Rank Approximation 2026

- Built end-to-end PyTorch pipeline for inverse RoPE de-rotation, truncated SVD, and progressive online recompression.
- Conducted 7-configuration evaluation sweep over 1,602 LongBench samples tracking ROUGE-L, perplexity, latency, and GPU memory usage.
- Quantified 29.5% latency overhead of Python-loop SVD and explored Triton-based batched randomised SVD optimisation.
- Demonstrated compatibility between activation-level compression and 4-bit NF4 quantised inference.

OCR + LLM Reconstruction Pipeline (Multimodal NLU) 2026

- Built multimodal document reconstruction pipeline combining image preprocessing, OCR, and LLM-based reasoning.
- Explored latency-quality trade-offs between local quantised inference and API-based models.
- Analysed hallucination behaviour and output stability across prompting strategies.

Transformer Encoder-Decoder from Scratch 2025

- Implemented multi-head self-attention, masking, positional encoding, and autoregressive decoding in PyTorch.

- Studied gradient flow, training stability, and architectural behaviour compared with RNN/LSTM baselines.

Raga Classification — Audio Deep Learning 2025

- Built 68-class audio classification system for Indian classical ragas using mel-spectrogram and MFCC representations.
- Achieved 68% classification accuracy while studying non-Western musical feature extraction challenges.

YOLO-Based Rockfall Detection 2025

- Trained YOLO object detection model for safety-critical rockfall detection on limited datasets.
- Analysed precision-recall trade-offs under varying IoU thresholds.

Industry Experience

Daten & Wissen Pvt. Ltd. Thane, India
AI/ML Intern 2025

- Fine-tuned PaddleOCR models for license plate detection and recognition under noisy real-world conditions.
- Improved robustness through geometry correction, adaptive filtering, and contrast normalisation.
- Evaluated segmentation architectures including U-Net, SAM, and ResNet backbones for production-oriented latency-accuracy trade-offs.
- Conducted Optuna hyperparameter optimisation sweeps and analysed convergence stability.
- Built end-to-end document understanding pipelines using PyTorch and TensorFlow.

Technical Skills

Languages: Python, C, C++, Java, SQL, CUDA C, JavaScript
ML/DL: PyTorch, TensorFlow, Hugging Face Transformers, PEFT/LoRA, BitsAndBytes, PaddleOCR, YOLO, Ollama
Systems: CUDA kernel development, cuBLAS, Triton, GPU profiling, model quantisation
Tools: Git, GitHub, Kaggle, Jupyter, VS Code, Optuna, LongBench, Google Colab